

Low Priority Process

```
root@dhanashree-Inspiron-5559:/media/dhanashree/New Volume/
Study/sem2/ProjectElective/Final Test# ./high
High process arrived at : 1522160421
Press any button to enter into critical section
```

```
High process waiting for lock start Time: 1522160451
High Priority Process started working at : 1522160451
My Priority is : 38
My Priority is : 38
endTime: 1522160456
Process completed
```

```
root@dhanashree-Inspiron-5559:/media/dhanashree/New Volume/
Study/sem2/ProjectElective/Final Test#
```

High Priority Process

```
root@dhanashree-Inspiron-5559:/media/dhanashree/New Volume/
Study/sem2/ProjectElective/Final Test# ./med
Med Priority Process arrived at: 1522160425
Press any button to start work...
start Time: 1522160431
Med Priority Process started working at : 1522160431
My Priority is : 28
My Priority is : 28
end Time: 1522160451
Process completed
root@dhanashree-Inspiron-5559:/media/dhanashree/New Volume/
Study/sem2/ProjectElective/Final Test#
```

Medium Priority Process

From the above code, we can see that *Lo* took extra time than the delay mentioned to come out of the critical section, i.e., almost along with the *Med* process. And then *Hi* started its work. This is due to all the processes running on the same processor, and because semaphore doesn't allow priority inheritance.

When processes *Lo* and *Hi* use PI-Futex: All the code for *fu.c*, *PriInhLow.c*, *PriInhHigh.c* and *PriInhMed.c* is compiled and their executables are stored as *fu*, *low*, *high* and *med* respectively. First *fu* is executed and then all the remaining run almost together. In this case, *Lo* and *Hi* use futex for the locking mechanism.

```
root@dhanashree-Inspiron-5559:/media/dhanashree/New Volume/
Study/sem2/ProjectElective/Final Test# ./low
My pid: 677
Press any button to enter into critical section
```

```
Aquired lock...
futex_add : 677
Low Priority Process started working at : 1522220815
Priority : 20
```

Low Priority Process

```
root@dhanashree-Inspiron-5559:/media/dhanashree/New Volume/
Study/sem2/ProjectElective/Final Test# ./high
High Priority Process arrived at 1522220809 and waiting for
lock
```

```
My pid: 679
Press any button to enter into critical section
```

```
futex_add : 677
Waiting for lock...
```

High Priority Process

```
root@dhanashree-Inspiron-5559:/media/dhanashree/New Volume/
Study/sem2/ProjectElective/Final Test# ./med
Medium Priority Process arrived at : 1522220813
Press any button to enter into critical section
```

Medium Priority Process

So, here, when *Lo* has futex and is performing the CS task, *Hi* is waiting for futex lock. And when *Med* arrives, it can't pre-empt *Lo*.

```
root@dhanashree-Inspiron-5559:/media/dhanashree/New Volume/
Study/sem2/ProjectElective/Final Test# ./low
My pid: 677
Press any button to enter into critical section
```

```
Aquired lock...
futex_add : 677
Low Priority Process started working at : 1522220815
Priority : 20
Priority : 20
Work completed at : 1522220830
```

Low Priority Process

```
root@dhanashree-Inspiron-5559:/media/dhanashree/New Volume/
Study/sem2/ProjectElective/Final Test# ./high
High Priority Process arrived at 1522220809 and waiting for
lock
My pid: 679
Press any button to enter into critical section
```

```
futex_add : 677
Waiting for lock...
High Priority Process started working at : 1522220830
Priority : 38
```

High Priority Process

```
root@dhanashree-Inspiron-5559:/media/dhanashree/New Volume/
```